Interpretability-Guided Circuit Routing for Hindi–English LoRA Fine-Tuning

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Abstract

Standard Uniform LoRA spreads its budget evenly over a model. We ask whether causal importance scores from activation patching can tell us where to place that budget when adapting to Hindi - English text. We use the Qwen2.5 1.5B model for our analysis. We patch all 336 attention heads and all 28 MLP layers between English, Romanized Hindi and Devanagari Hindi versions of concepts drawn from Hindi WordNet and IndoUKC. We use two sets: a pilot set and a larger 120 concept set, which give broadly similar localisation results. We then use the scores to gate the LoRA update itself on heads, MLP layers or both, and compare with random gates of the same size and with uniform LoRA on SAIL-2017 Hinglish sentiment. We call this gated LoRA update - Circuit Routing Adapter (CRA). We conclude by showing that our CRA showed no reliable advantage over random routing for any adapter.

1 Introduction

We observe in general life that language models trained mostly on English still work reasonably well in Hindi. Hindi, which is often written in Devanagari / Roman script and sometimes even combined with english to give what is informally called : Hinglish. Adapting a model to such text is usually done with low rank adapters (Hu et al., 2021), which spread a fixed budget evenly over the model. We reason that if we knew where language related computation happens, we might be able to efficently spend that budget.

Interpretability work suggests where to look. A few attention heads matter for each language (Liu et al., 2026), and word level translation relies on fewer than 5% of heads together with specific MLP layers (Zhang et al., 2025). Some of these findings have been turned into adaptation methods (Yin et al., 2024; Liu et al., 2026; Zhang et al., 2025), although heads chosen for one task can be worse on another (Yin et al., 2024). The general idea of using interp to guide adaptation is therefore not new. Our question is narrower: whether causal patching scores, used to gate the LoRA update and compared with budget matched random gates, give any measurable gain.

We run activation patching on the attention heads and MLP layers of Qwen2.5 1.5B, using concepts sampled from Hindi lexical resources across meaning and part of speech. We then check whether heads act independently, as a ranking of single heads is a sound guide to sets of heads only if they do. Finally, we use the scores to route LoRA updates to the highest scoring heads, MLP layers, or both, and compare each with random choices of the same size and with uniform LoRA.

Method.

- A routing scheme in which fixed, non trainable masks derived from patching scores/ random controls scale the LoRA update, leaving the frozen base weights untouched, with random controls matched on the number of active units (Section 3).

Empirical findings.

- Head level and MLP layer causal localisation of language and script in Qwen2.5 1.5B, with famil -wise tests (Section 5.1).
- Evidence that heads interact (Section 5.2).
- A negative result, i.e across three adapter types and three training seeds, routing by patching scores showed no reliable advantage over matched random routing and fell slightly below uniform LoRA (Section 5.3).

- Robustness checks of the localisation findings on a pilot set and a larger 120 concept set that overlaps with the main one.

2 Related Work

Our literature review revealed to us some useful insights, which we carried forward in our project. Activation patching restores a component's clean activation in a corrupted run and measures how much of the output returns. It grew out of causal mediation analysis (Vig et al., 2020) and has been used to localise factual recall (Meng et al., 2022) and to recover circuits. (Wang et al., 2023). Conmy et al. (2023) automate the search. Results depend on how the corrupted prompt is built and on the metric (Zhang and Nanda, 2024).

Feed-forward layers act as key value memories whose outputs promote specific tokens (Geva et al., 2021), and causal tracing places factual recall in middle-layer MLPs (Meng et al., 2022). For translation, Zhang et al. (2025) find that heads extract language and position features which specific MLPs then integrate. We therefore patch MLP layers alongside heads.

Patching one component at a time assumes their effects add up. However, they need not, ie ablating one component can make others compensate (McGrath et al., 2023), and Held and Yang (2023) estimate head contributions with Shapley values, which average over subsets of the other heads. Heimersheim and Nanda (2024) distinguish restoring components into a corrupted run (sufficiency) from corrupting them in a clean run (necessity), and discuss patching several at once; circuit work patches paths and sets of heads jointly (Wang et al., 2023; Conmy et al., 2023). We measure joint effects directly, using sets of heads of growing size and every pair in a pre selected shortlist, alongside family wise tests.

In translation encoders a few heads do most of the work (Voita et al., 2019). In multilingual models some heads interfere across languages (Held and Yang, 2023) and language specific subnetworks reduce this (Choenni et al., 2023). In decoder models language specific neurons have been found (Tang et al., 2024; Kojima et al., 2024), intermediate layers of Llama-2 decode most readily into English (Wendler et al., 2024), and Romanized forms of Indic words appear in intermediate layers before native script (Saji et al., 2025).

LoFiT tunes offsets on 3-10% of heads (Yin et al., 2024), Liu et al. (2026) train a gate over language heads, Zhang et al. (2025) fine-tune translation heads, Wang et al. (2025) use circuits to set LoRA ranks, and Mondal et al. (2025) restrict LoRA to language-specific neurons without gains. Our setup differs from these in the details of the test, but not in the basic idea. We gate the LoRA update by patching scores, and compare against random gates with the same number of active units. However, we do not quantise the base model. The scoring task also differs from the training task: we score on word retrieval and train on sentiment. SAIL-2017 (Patra et al., 2018) is part of GLUECoS (Khanuja et al., 2020).

3 Problem Statement and Hypotheses

Setting.

- **Model:** L=28 layers, each with H=12 query attention heads (N=336 head positions) and one MLP block.
- **Components:** head (l, h) writes a 128-dimensional vector (its slice of the attention output). MLP block l writes one vector into the residual stream. Both are taken at the last prompt token.
- **Concepts:** each concept k has a prompt p_c^k and a one-word target y_c^k in three conditions $c \in \{\text{en}, \text{hi_deva}, \text{hi_latn}\}$.
- **Contrasts:** *language* (en vs. hi_latn, script fixed), *script* (hi_deva vs. hi_latn, language fixed), *both* (en vs. hi_deva).

Score, contrast and recovery. The score of a target is its mean per-token log-probability:

$$\boxed{s(y \mid p) = \frac{1}{|y|} \sum_{i=1}^{|y|} \log P(y_i \mid p, y_{<i})} \quad (1)$$

For a source condition a and target condition b, the contrast on a prompt p is

$$\boxed{m^k(p) = s(y_a^k \mid p) - s(y_b^k \mid p)} \quad (2)$$

Let $m^k_{\text{clean}} = m^k(p_a)$ and $m^k_{\text{corr}} = m^k(p_b)$, and let m^k_S be the contrast on p_b when the components in a set S take their values from p_a. The recovery of S is

$$\boxed{R^k(S) = \frac{m^k_S - m^k_{\text{corr}}}{m^k_{\text{clean}} - m^k_{\text{corr}}}} \quad (3)$$

and $\bar{R}(S)$ is its mean over concepts. The interaction of heads A and B is

$$\boxed{I(A,B) = \bar{R}(\{A,B\}) - \bar{R}(A) - \bar{R}(B)} \quad (4)$$

Circuit Routing Adapter (CRA). A LoRA layer adds $\frac{\alpha}{r}BAx$ to the output of a frozen weight W_0x. The CRA multiplies only this update by a fixed mask, so the frozen base projection is never edited or pruned:

$$\boxed{h = W_0x + \frac{\alpha}{r}\Delta(x;G)} \quad (5)$$

where the form of Δ depends on the projection. We adapt `q_proj` and `o_proj` in every layer.

For `q_proj`, heads partition the output dimension, and the mask scales the corresponding output rows of the LoRA update, $\Delta = G_{\text{out}} \odot (BAx)$.

For `o_proj`, heads partition the input dimension, and the mask is applied to the input before the low-rank projection, $\Delta = BA(G_{\text{in}} \odot x)$. Both wrappers of a layer use the same head mask $g_{l,h}$.

For MLP routing, a single scalar gate g_l scales the LoRA update on the `down_proj` of layer l. The A and B factors are the only trainable parameters. The masks are not trained.

Head routing is global. Top-k selects the k highest signed scores over all N=336 head positions (layers $\times$ heads), not k heads per layer,

$$\boxed{g_u = \mathbb{1}\left[\bar{R}_u \geq \bar{R}_{(k)}\right]} \quad (6)$$

where $\bar{R}_{(k)}$ is the k**-th largest score**. The implementation takes the top k of the signed scores, so strongly negative scores are not selected because of their magnitude. MLP routing is at the level of whole layers, using the whole-layer MLP patching scores.

Its variants are: **uniform** (every layer has gate 1)//**top-**k (the k highest-scoring layers have gate 1)//**soft** (gate $\max(0, \bar{R}_l)/\max_{l'} \bar{R}_{l'}$)// **random** ($k$ layers chosen at random).

Layers with a zero MLP gate are not wrapped with an adapter. Random head routing opens the same number of heads as top-k, chosen at random.

Hypotheses.

H1 (localisation) A few heads and MLP layers must carry the language and script contrasts.

H2 (independence) Heads act additively, so single head scores must suffice to rank sets of heads.

H3 (routing) With the same number of active units, routing by our scores beats random routing on SAIL-2017 for (a) heads, (b) MLP layers and (c) heads and MLP layers together.

H4 (efficiency) A routed adapter comes close to uniform LoRA with fewer mask active adapter parameters

4 Method

4.1 Model and Setup

We use Qwen2.5-1.5B (Qwen Team, 2024) at a pinned revision. The model is loaded in FP16 without quantisation for both patching and Stage 2 training. Although the saved configuration files contain NF4 fields, quantisation is disabled in the Stage 2 loading code. All experiments were run on NVIDIA RTX 2080 Ti GPUs, except for one run described in Section 6. A single pipeline carries out the Stage 1 analysis, including validation, head and MLP patching, joint and pair patching, statistical tests, and report generation.

4.2 Concept Set

We constructed the concept set using Hindi WordNet and IndoUKC (Nair et al., 2022). After merging entries with the same meaning, we manually selected 50 concepts across 20 groups defined by semantic category and part of speech: 25 nouns, 15 verbs, 8 adjectives, and 2 adverbs. This follows the broad-coverage motivation of Multi-SimLex (Vulić et al., 2020) rather than restricting the set to common nouns. We ran the full set through the pipeline.

Each concept consists of a short definition and a request to produce the corresponding word. We wrote prompts in English, Romanized Hindi, and Devanagari Hindi, using the word itself as the target (Table 6, Appendix A). A concept is retained for a contrast when its target has sufficient probability under its corresponding prompt and the two conditions differ sufficiently. These thresholds were fixed in advance. The saved validation output reports 50 usable concepts for English–Devanagari,

48 for English–Romanized, and 45 for Devanagari–Romanized.

The primary Stage 1 analysis uses 39 concepts, selected from the 45 retained concepts using the prespecified validation thresholds: a log-probability floor of -8 and a contrast threshold of 0.5. Results using the 45 concepts retained under the looser thresholds applied during the run are presented in Section 6. Stage 2 uses the primary-set scores stored.

The run manifest records failures in validation and head analysis. Head analysis stopped when the observed mean contrasts differed by more than the hard-coded 15% tolerance from reference values inherited from an earlier concept set. We retained the observed measurements and analysis outputs. The validation log flags five concepts for rewriting or removal but does not document the failure further. The released code records the modifications to the working tree used for this run. The resulting concept set provides a targeted evaluation inventory rather than an exhaustive taxonomy of Hindi meanings. We did not conduct expert linguistic annotation.

4.3 Patching and Statistics

For each concept and contrast, we replace one component's output at the last prompt token in the target run with its corresponding output from the source run. The three directions are Romanized Hindi into English (language), Devanagari Hindi into Romanized Hindi (script), and Devanagari Hindi into English (both). We perform this intervention separately for all 336 attention heads and 28 MLP layers, then recovery is calculated. Concepts are the units of analysis.

We estimate uncertainty using percentile bootstrap intervals over concepts (Efron and Tibshirani, 1993). For multiple comparisons, we use a sign-flip permutation test based on the maximum statistic (Nichols and Holmes, 2002). The head analysis uses mean recovery as its statistic, whereas the MLP-layer and pair analyses use t-statistics to account for differences in spread. Each family comprises the 336 heads, 28 layers, or 1,225 shortlisted pairs tested in the corresponding analysis. We also perform a leave-one-concept-out analysis to measure how strongly each top head depends on any individual concept.

All bootstrap and permutation procedures use 2,000 resamples or permutations, with seed 0 and a family-wise significance level of 0.05. The head test is one-sided: each head's mean recovery is compared against the 95th percentile of the maximum head mean obtained from sign-flipped concept sets. For MLP layers and pairs, we use the maximum absolute t-statistic, giving two-sided tests. We report 95% percentile bootstrap intervals, with 0.025 in each tail. For the leave-one-concept-out analysis, we record the largest change in each top head's mean recovery after removing one concept, expressed as a fraction of the original mean.

With 336 heads, individual positive intervals are not sufficient to establish significance. Under a global null, a one-sided threshold of 0.025 would yield an expected 8.4 heads above the threshold. We therefore use the family-wise permutation tests to assess evidence across the full set of heads.

4.4 Joint Patching and Pair Selection

Sufficiency measures the recovery obtained by patching a set of heads into the corrupted run simultaneously (Eq. 3). *Necessity* measures the fraction of the contrast removed when a head's corrupted output is patched into the clean run. We patch the top-k heads ranked by sufficiency jointly for every k from 1 to 336, comparing their recovery with the sum of their individual effects and with the recovery from k randomly selected heads.

Evaluating all $\binom{336}{2} = 56{,}280$ head pairs would require approximately 60 GPU-hours. We therefore construct a shortlist using single-head scores before measuring any pairs:

1. Rank all 336 heads separately by sufficiency and necessity.
2. Alternate between the two rankings, selecting the next-best head from each and skipping duplicates, until 50 heads have been selected. The resulting shortlist contains 27 heads from the sufficiency ranking and 23 from the necessity ranking.
3. Patch all pairs among the 50 selected heads, yielding 1,225 pairs, and remeasure the individual heads in the same run.

Including necessity in the selection procedure captures heads that substantially affect the contrast when removed, even if their individual recovery is small. We also evaluate 190 randomly selected pairs outside the shortlist as a control. Family-wise tests are applied across the 1,225 shortlisted pairs. A pair is classified as *robust* when it passes the

t-based test using both single-head estimates: the original trace and the remeasurement.

4.5 Stage 2: Routing LoRA on SAIL-2017

We evaluate routing on the Hinglish sentiment task from SAIL-2017 (Patra et al., 2018), which contains 10,068 training, 1,260 validation, and 1,261 test examples across three classes. Classification is performed through next-token prediction: the model scores each label word, and the highest-scoring label is selected. We report macro-F1 on the validation split.

Routing scores are the language-contrast recoveries measured on the 39 primary concepts and stored in the `ling39` score files. Our baselines include uniform LoRA on the query and output projections. For routed adapters, we compare top-k selection with random selection at the same budget: 112 or 34 of the 336 attention-head positions, three MLP layers using `down_proj`, or heads and MLP layers together. Soft MLP routing provides an additional comparison.

All arms use LoRA rank 8 and $\alpha = 16$, giving a scaling factor of $\alpha/r = 2$, with dropout 0.05. The LoRA factors are stored in FP32, while the frozen base model remains in FP16. Training uses a learning rate of 10^{-4}, weight decay of 0.01, 600 steps, and 30 warmup steps. We set the maximum sequence length to 256 and use a batch size of 4 with gradient accumulation over 8 steps, giving an effective batch size of 32. Gradient clipping is set to 1.0 (Appendix B).

The principal arms use seeds 0, 1, and 2. Each seed fixes the data order and initialisation. For random-routing arms, it also fixes the selected units. Results for arms sharing a seed are therefore paired.

4.6 Adapter Parameter Accounting

The attention wrappers allocate full LoRA A and B tensors, while the routing masks determine which dimensions can receive nonzero gradients. We report two parameter counts. *Mask-active adapter parameters*, computed by the implementation's `effective_params()` function, count the dimensions capable of receiving nonzero gradients under the mask. *Allocated trainable adapter parameters* count all existing LoRA parameters with `requires_grad` enabled, excluding the frozen base model.

Zero-gated MLP layers are not wrapped and allocate no adapter parameters. By contrast, every wrapped attention projection allocates its full LoRA factors, regardless of how many heads are active under the mask. Table 1 reports both counts. The training logs record the allocated counts, including 1,376,256 for each attention arm and 1,628,160 for `HM-topk`. These values are not included in the saved metrics files and can also be derived from the wrapper dimensions. Mask-active counts are stored for each run.

Configuration	Mask-active	Allocated
H-uniform (all heads)	1,376,256	1,376,256
H-topk112	917,504	1,376,256
H-random112	868,352–917,504	1,376,256
H-topk34	512,000	1,376,256
H-random34	512,000–561,152	1,376,256
HM-topk	1,169,408	1,628,160
HM-random	1,120,256–1,169,408	1,628,160
M-topk3, M-random3	251,904	251,904
M-soft (17 layers)	1,427,456	1,427,456

Table 1: Adapter parameter counts. Mask-active parameters are dimensions able to receive nonzero gradients under the routing mask. Allocated parameters are trainable LoRA parameters that exist, excluding the frozen base model. Random arms show the range over seeds 0–2. The 34-head and 112-head attention adapters have the same allocated size.

For top-k routing, the mask-active count is fixed by the selected budget. Random routing can produce different counts across seeds because the distribution of selected heads across layers affects the number of active dimensions in each layer .

5 Results

5.1 Where Language and Script Are Carried (H1)

MLP layers recover more than any single head (Table 3). On the language contrast 15 of 28 layers are significant, and the strongest are L17 (0.131), L22 (0.117) and L23 (0.110), in the same late layers as the leading heads. A whole MLP block is a much larger unit than a head, so the two are not compared one-to-one. H1 is supported for both component types on this concept set.

5.2 Heads Are Not Independent (H2)

Joint patching. Patching the top heads together recovers more than the sum of their single effects for small sets: 0.587 against 0.495 at k=20 (Figure 2; Table 8). For large sets the effect saturates: at k=100 the joint recovery is 0.949 against a sum of 1.053, while 100 random heads recover 0.040. All 336 heads together recover 0.881. A head's

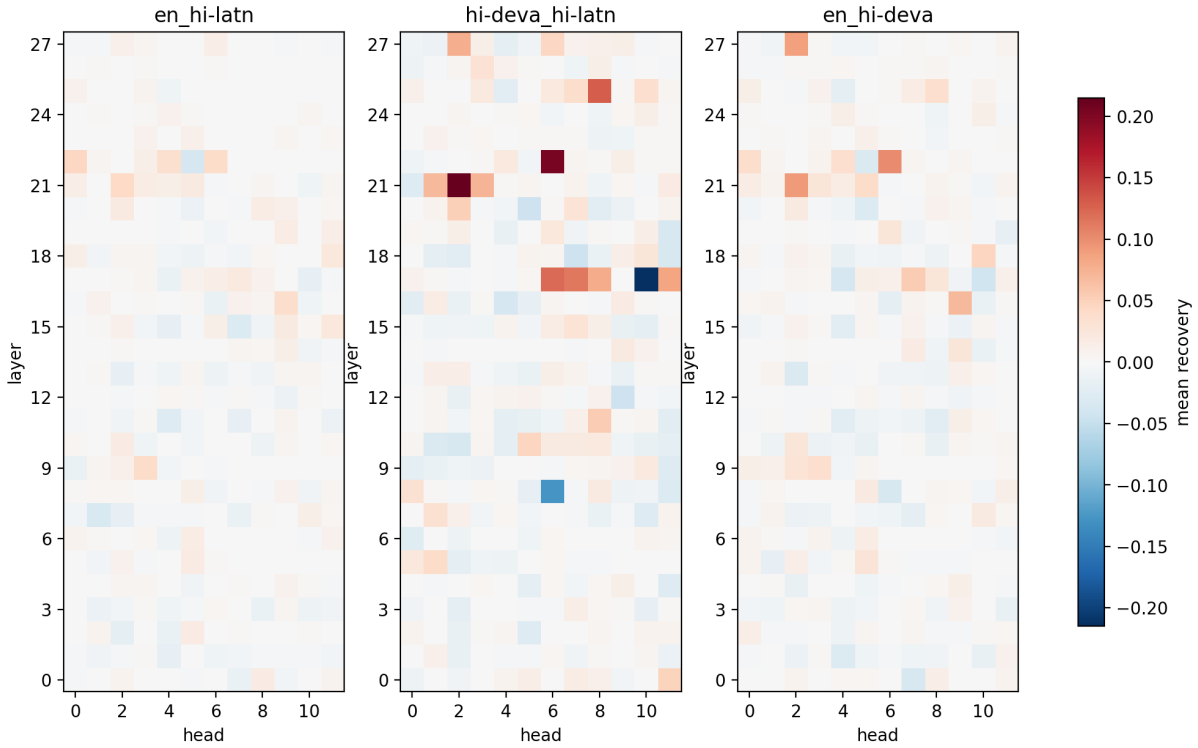


Figure 1: Mean recovery of each head (layer × head) for the three contrasts, 39 concepts of the linguistic set.

Contrast	Contrast size	τ	Significant	CI > 0 (chance)	Top three heads (recovery)
Language (En–Rom.)	11.50	0.0173	14	116 (8.4)	L22H0 0.044, L21H2 0.043, L22H6 0.039
Script (Dev.–Rom.)	2.20	0.0971	5	122 (8.4)	L21H2 0.303, L22H6 0.204, L25H8 0.130
Both (En–Dev.)	8.39	0.0331	11	118 (8.4)	L22H6 0.101, L21H2 0.092, L27H2 0.087

Table 2: Single-head patching on the linguistic set (39 concepts). Mean contrast in log-probability units; τ: family-wise threshold on mean recovery over 336 heads; Significant: heads above τ; CI > 0: heads whose 95% bootstrap interval excludes zero, with the number expected by chance. The script contrast is small, which inflates its recoveries.

Contrast	τ_t	Sig.	Top 3 layers (recovery)
Language	3.14	15	L17 0.131, L22 0.117, L23 0.110
Script	3.03	18	L22 0.318, L9 0.217, L25 0.193
Both	3.18	17	L27 0.175, L25 0.172, L23 0.171

Table 3: MLP-output patching per layer (28 layers). τ_t: max-$|t|$ family-wise threshold; Sig.: number of significant layers.

necessity and sufficiency correlate only moderately (0.62); L21H0, for example, removes 0.070 of the contrast when corrupted but recovers only 0.015 alone.

Pairs. Of the 1,225 shortlisted pairs (all pairs among 50 heads), 137 are robust. 123 of them lie outside the range of the random control pairs ([−0.0014, +0.0012]) and 17 are negative, meaning redundant. The strongest are L21H2+L21H3 (I=0.0078, t=11.2), L17H8+L21H2 and L22H0+L22H4. The test on the mean finds only 2 pairs, because a few very variable pairs set its threshold. The t-based test standardises each pair and is the one we use. H2 is not supported. Single head scores do not determine the effect of a set.

5.3 Routing LoRA by Scores (H3, H4)

Table 4 gives the Stage 2 results. Zero-shot, the base model reaches macro-F1 0.319. Uniform LoRA reaches 0.604. Mean macro-F1 over three seeds is 0.592 for top-k heads and 0.594 for random heads at k=112; 0.576 and 0.583 at k=34; 0.573 and 0.575 for three MLP layers; and 0.590 and 0.592 for heads plus MLP layers. Confidence intervals below are percentile bootstrap intervals over validation examples applied to the paired dif-

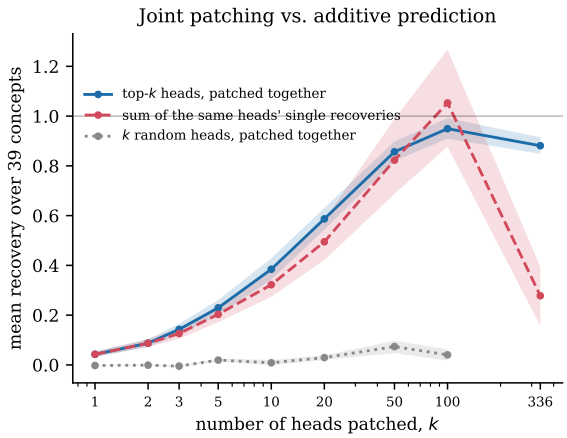


Figure 2: Joint patching on the language contrast: the top-k heads patched together, the sum of their single-head recoveries, and k random heads.

Arm	Seeds	Macro-F1	Trainable
Uniform LoRA	3	0.604 ± 0.005	1,376,256
H-top-k	3	0.592 ± 0.007	917,504
H-random	3	0.594 ± 0.003	868,352–917,504
M-top-k	3	0.573 ± 0.003	251,904
M-random	3	0.575 ± 0.001	251,904
HM-top-k	3	0.590 ± 0.004	1,169,408
HM-random	3	0.592 ± 0.005	1,120,256–1,169,408

Table 4: SAIL-2017 validation macro-F1 (mean $\pm$ sample standard deviation over three seeds). H: heads, M: MLP layers, HM: both; uniform: all attention heads (standard LoRA on *q*/*o*); top-k / random: routed by score / at random. Trainable: effective number of LoRA parameters receiving gradients. Ranges indicate variation in effective parameter counts across random-routing seeds.

ferences. They capture evaluation-set sampling, not variation between training runs; only the three seeds speak to the latter, and three is few. The bootstrap uses 2,000 resamples with seed 0; each resample draws validation examples with replacement and recomputes the macro-F1 of both arms on the same examples. The per-seed predictions are concatenated before resampling, so one interval summarises the three seeds jointly. The per-seed differences are reported separately.

H3: routing against random. We find no evidence that top-k routing outperforms random routing under our decision rule. For heads at k=112, the differences are close to zero in all three seeds (-0.000, $+0.000$, and -0.007), with a mean of -0.002 and an interval of $[-0.009, +0.004]$. The MLP comparison is similarly inconclusive: the seed-wise differences are -0.002, $+0.001$, and -0.002, giving a mean of -0.001 and an interval of $[-0.011, +0.009]$. Combining heads and MLP layers (H3c) does not change the picture. The mean difference is -0.002, and the interval remains compatible with zero ($[-0.009, +0.005]$).

At k=34, top-k heads score below random routing in each seed, by 0.012, 0.005, and 0.002, respectively. However, the interval $[-0.014, +0.000]$ reaches zero. We therefore find no clear advantage for either method at this setting. The intervals containing zero indicate that the experiment does not establish a difference; they should not be interpreted as evidence that the methods are equivalent. Adding MLP layers to the top-k heads also has little effect, with a mean difference of -0.002 and an interval of $[-0.007, +0.003]$.

H4: efficiency. The results provide only partial support for H4. With 112 selected heads, top-k routing obtains a macro-F1 of 0.592, compared with 0.604 for uniform LoRA. It falls short in all three seeds, by 0.013, 0.006, and 0.016, respectively. The mean difference is -0.012, with an interval of $[-0.020, -0.004]$. Thus, the performance gap is small in absolute terms, but the results do not support describing the two methods as equivalent.

The routed adapter has 917,504 mask-active parameters, compared with 1,376,256 for uniform LoRA (Table 1). This is roughly two thirds of the uniform LoRA count. The reduction comes partly from the head-specific dimensions: according to the accounting function, selecting 112 heads activates 229,376 such dimensions, compared with 688,128 when all heads are active. The other dimensions remain active whenever a layer has at least one selected head.

This reduction in mask-active parameters does not translate into a smaller adapter. All attention arms allocate the same 1,376,256 trainable parameters, so routing does not reduce the stored adapter size or its memory footprint.

The random baselines also have the same number of selected heads as top-k, although their mask-active counts vary across seeds. At k=112, random routing uses 868,352 parameters in seed 0 and 917,504 in seeds 1 and 2; top-k uses 917,504 in every seed. At k=34, the corresponding random counts are 512,000, 536,576, and 561,152, while top-k uses 512,000. The comparison is therefore matched in selected heads and allocated parameters, but not exactly in mask-active parameters. In particular, top-k uses no more mask-active param-

eters than random at k=34, and uses the same or more at k=112. This difference in active parameter count does not account for the absence of a routing advantage.

Performance and mask-active parameters. Across the evaluated arms, macro-F1 tends to increase with the number of mask-active adapter parameters. It is approximately 0.573 for the three-MLP-layer arm (252k parameters), 0.58 at 512k, 0.59 for arms between 0.9M and 1.2M, and 0.604 for uniform LoRA at 1.38M. This is a pattern across the configurations we evaluated, rather than evidence that increasing the active parameter count causes better performance. In particular, all attention arms allocate the same number of trainable parameters, so the pattern should not be interpreted as a direct capacity comparison.

6 Ablations and Sensitivity

The pilot and 120-concept sets below are robustness and sensitivity analyses. They are not independent replications: the 120-concept set shares concepts with the main study, and the sets were developed in sequence by the same authors.

Looser validation thresholds. During the run the validation thresholds were loosened, admitting 45 of 50 concepts. We treat the 39 that pass the thresholds fixed in advance as primary and recompute everything for the 45 from the same per-concept outputs (Table 5, third column). The leading heads, the joint-patching curve, the pair counts and the MLP layers barely change.

Pilot concept set. Before building the linguistic set we used a pilot of 38 concrete concepts in nine categories (colour, animal, body, food and so on), written as fill-in sentences and reported by the authors as checked by a native speaker. Its contrasts are larger, but its single heads recover less (Table 5, first column): 63 heads have intervals above zero and four pass the family-wise test, of which one (L0H10) depends on a single concept. L21H2 and L22H6 lead in both sets; L27H6, third in the pilot, ranks 24th in the linguistic set. The pilot's pair test used a 100-head shortlist (4,950 pairs) and gave 11 robust pairs. Of these, 10 are in the linguistic shortlist and 9 of the 10 are robust again with the same sign, even though that shortlist was chosen from the linguistic set's own scores. On the pilot the t-based test was added after the mean-based test found a single unstable pair; on the linguistic set both were fixed in advance.

Larger concept set. We also ran the same pipeline on a second set of 120 concepts (six per group), whose Hindi and Romanized texts were checked by one of the authors. Of the 120 concepts, 95 are new and 25 share an id with the main set (all reworded). Ninety-nine pass the thresholds fixed in advance, and we report these. The picture largely holds. L21H2 and L22H6 remain among the leading heads, although L16H9 leads here and was not in the main set's top three. In all, 22 heads pass the family-wise test, and the strongest MLP layers are again L22, L17 and L23. Joint patching of the top 20 heads recovers 0.654 against a sum of 0.528. Of the 77 robust pairs of the main set that this set's 35-head shortlist could test, 76 are robust again with the same sign. The caveats are these. The overlap in concepts means this is a sensitivity analysis and not an independent test. Three concepts (E300-028, E300-041, E300-185) were reworded using the validator's scores, which is a small selection effect. The run used a different GPU (A100). The smaller shortlist means pair counts do not compare directly with the main set.

Secondary Stage 2 arms. Soft MLP routing, uniform LoRA on MLP layers and heads and MLP layers at k=34 were run as extras, with one seed each. They appear in Table 4, play no part in deciding H3, and carry less evidential weight than the three-seed arms. An earlier single-seed comparison with pilot scores also showed no clear difference between top-k and random heads.

7 Discussion

Stage 1 gives a broadly consistent picture across the three concept sets: a small group of late heads, chiefly L21H2 and L22H6, carries the language contrast; late MLP layers carry more; and heads combine non-additively, with many of the same pairs interacting in each set. The sets overlap and share authors and pipeline, so we read this as a sensitivity check and not as three separate confirmations. A ranking of single heads is a coarse guide to where an update should go.

Stage 2 gives no support for routing by patching scores in this setting. H3a, H3b and H3c all fail the decision rule, including H3c, which was our main goal, and H4 is only partly met. The finding applies to this model, this task, this training

	Pilot	Linguistic set	
		primary	sensitivity
Concepts (validation rule)	38 (-8, 0.5)	39 (-8, 0.5)	45 (-10, 0.3)
Mean contrast: lang. / script / both	13.3 / 6.2 / 16.2	11.5 / 2.2 / 8.4	11.7 / 2.3 / 8.3
FWER heads (language)	4	14	18
Heads with CI > 0 (chance 8.4)	63	116	117
Top two heads	L21H2 0.025, L27H6 0.023	L22H0 0.044, L21H2 0.043	L21H2 0.045, L22H0 0.044
Joint / sum, k=20	0.38 / 0.24	0.59 / 0.49	0.60 / 0.50
Joint / sum / random, k=100	0.97 / 0.45 / 0.17	0.95 / 1.05 / 0.04	0.95 / 1.06 / 0.04
All 336 heads together	0.97	0.88	0.89
Shortlist pairs	4,950	1,225	1,225
Robust pairs (of which negative)	11 (0)	137 (17)	166 (22)
beyond random-pair range	11	123	147
Pilot robust pairs replicated	–	9 of 10	9 of 10
Spearman(I, product of singles)	0.05	0.36	0.37
MLP layers significant (language)	5	15	17
Top MLP layer (language)	L22 0.090	L17 0.131	L17 0.127

Table 5: Pilot set against the linguistically sampled set. Primary: the concepts passing the validation thresholds fixed before the study; sensitivity: the 45 concepts passing the loosened thresholds used during the run. Pairs: 100-head shortlist for the pilot, 50-head shortlist for the new set, chosen from each set's own scores. Robust: significant under max-$|t|$ with both single-head baselines. All on the language contrast unless stated.

budget and this routing implementation. It does not show that causal routing cannot help, and non-significant differences do not show equivalence. We did not test why routing failed, so the following are hypotheses. First, the scores come from word retrieval, while sentiment is a comprehension task, so patching importance may not predict where fine-tuning is useful. Intervention scores measure what a component does in the clean model, not how much an update to it helps. Second, Stage 1 showed that heads interact, and routing by single-head scores ignores that. Third, with this much budget almost any placement may be enough. The pattern that F1 tracks the number of mask-active parameters more than which parameters are active fits the third idea but does not establish it. Finally, our top-k rule ranks signed scores, so strongly negative heads are never selected; a magnitude-based rule might behave differently.

8 Conclusion

We localised Hindi–English language and script computation in Qwen2.5-1.5B by patching heads and MLP layers, found a small group of late heads and stronger late MLP layers, showed that heads interact, and found broadly similar patterns on a pilot set and an overlapping 120-concept set. Gating LoRA updates with these scores did not show a reliable advantage over matched random gates on SAIL-2017 Hinglish sentiment, for heads, MLP layers or both, and the routed head adapter stayed slightly below uniform LoRA. The mask-active parameter count of routed adapters is lower, but their allocated size is not.

Limitations

The linguistic set has 39 concepts in groups of two or three, too few to compare heads across linguistic features, and its concepts were chosen by hand for familiarity and ease of prompting, which may favour concepts the model finds easy. Results may be sensitive to concept wording and translation quality. Its validation thresholds were loosened during the run, which is why we treat the pre-specified subset as primary. The Hindi and Romanized texts of the 120-concept set were checked by one of the authors, not by an independent reviewer. These texts were verified by one co-author who is a native speaker of Hindi (Satyam Arora); this is an author check and not an independent annotation. The 120-concept set ran on a different GPU (A100) and includes three concepts reworded with the validator's scores, and the three concept sets overlap. All patching is at the last prompt token and on production (word retrieval) behaviour, while Stage 2 is a comprehension task, so Stage 1 importance may not predict fine-tuning utility. The pair test covers a shortlist of 50 heads (2.2% of all pairs) on the language contrast only, so other pairs are untested, not ruled out. MLP routing is at the level of whole layers, not neurons. In the pilot, head patching and MLP patching were separate runs, and its single-head trace and pair runs used different GPU models.

Stage 2 uses one model, one task, the validation split only, and three training seeds for the principal arms. Bootstrap intervals resample evaluation examples and do not capture variation between training runs, so with three seeds they are indicative only. Soft routing, uniform LoRA on MLP layers and the 34-head heads-and-MLP arms have one seed and are exploratory, and soft routing is not size-matched. Comparing heads-and-MLP against heads alone confounds the choice of units with parameter count, since the MLP layers add parameters. Top-k selection uses signed scores and ignores strongly negative ones. Mask-active counts do not measure adapter size: all attention adapters allocate the same LoRA parameters. We have no evidence on other languages, model families or tasks. The first four runs evaluated at steps 200, 400 and 600 and the rest only at the end.

What the study does establish is narrower. Patching scores localise language and script contrasts to a small set of late components, and routing by these scores gave no detectable gain over matched random routing under a controlled comparison. Open directions include neuron-level MLP patching, routing by score magnitude so that strongly negative heads are kept, more training seeds and smaller budgets (where placement may matter more), patching at all positions, scoring on a task closer to the training task, and more concepts per linguistic group.

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Concept	Stratum	Prompt and target (bold)
E300-001	Animals	A common pet that barks and guards the house. The English word for this is: **dog** Ek aam paaltu jaanvar jo bhaunkta hai aur ghar ki rakhvaali karta hai. Iske liye Hindi shabd hai: **kutta** एक आम पालतू जानवर जो भौंकता है और घर की रखवाली करता है। इसके लिए हिंदी शब्द है: **कुत्ता**
E300-151	Motion and posture	To move forward on foot, one step after another. The English word for this is: **walk** Pairon se ek-ek kadam rakhkar aage badhna. Iske liye Hindi shabd hai: **chalna** पैरों से एक-एक क़दम रखकर आगे बढ़ना। इसके लिए हिंदी शब्द है: **चलना**
E300-241	Colour and sensory properties	The color of ripe tomatoes and vermilion. The English word for this is: **red** Pake tamaatar aur sindoor ka rang. Iske liye Hindi shabd hai: **laal** पके टमाटर और सिंदूर का रंग। इसके लिए हिंदी शब्द है: **लाल**

Table 6: Three concepts from the linguistic set (a noun, a verb and an adjective) in English, Romanized Hindi and Devanagari Hindi.

A Concept Examples

Table 6 shows three concepts of the linguistic set (a noun, a verb and an adjective) in the three conditions.

B Training Settings

Setting	Value
Base model	Qwen2.5-1.5B, fp16
LoRA rank / α / dropout	8 / 16 / 0.05
Modules	q, o; MLP down proj.
Steps / effective batch	600 / 32
Learning rate	10^{-4}, warm-up 30, cosine
Max. length / weight decay	256 / 0.01
Seeds	0, 1, 2 (soft: 0)
Hardware	NVIDIA RTX 2080 Ti

Table 7: Stage 2 training settings, shared by all arms.

C Additional Results

k	Joint	Sum	Rand.	Joint−sum [95% CI]
1	0.043	0.043	−0.002	+0.000 [0.000, 0.000]
2	0.087	0.087	−0.001	+0.000 [−0.001, 0.002]
3	0.143	0.126	−0.005	+0.016 [0.012, 0.021]
5	0.229	0.203	0.019	+0.026 [0.017, 0.035]
10	0.385	0.322	0.009	+0.062 [0.042, 0.081]
20	0.587	0.495	0.029	+0.092 [0.040, 0.141]
50	0.856	0.823	0.074	+0.034 [−0.106, 0.152]
100	0.949	1.053	0.040	−0.104 [−0.293, 0.049]
336	0.881	0.278	–	+0.603 [0.487, 0.719]

Table 8: Joint patching, language contrast, linguistic set: top-k heads patched together (Joint), the sum of their single-head recoveries (Sum), and k random heads (Rand.).

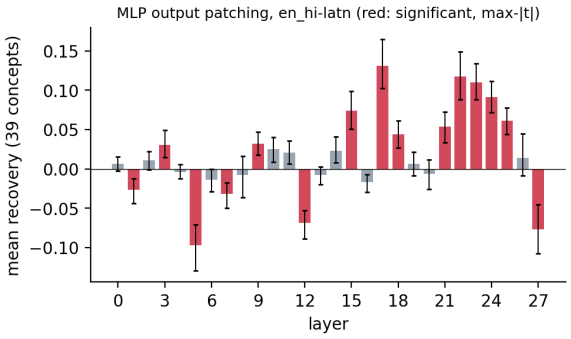


Figure 3: MLP-layer recovery, language contrast, linguistic set. Whiskers: 95% bootstrap intervals; red: significant (max-$|t|$).

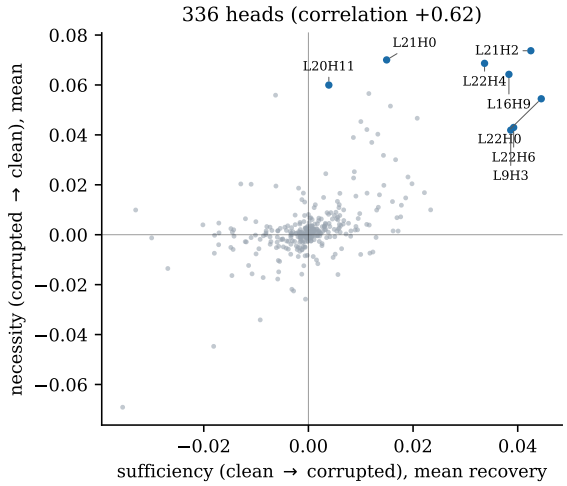


Figure 4: Necessity against sufficiency for each of the 336 heads, language contrast, linguistic set.

D Team Contributions

Balasubramanian K: the head-interaction analysis (joint and pair patching) and CRA.
Daniel Ashish Abraham: Stage 1 run, activation patching and the theoretical framing.
Jawed Mohammed Khan: MLP-layer patching.
Satyam Arora: concept-set design, the Stage 1 run on the 120-concept set and the check of its Hindi and Romanized texts.